

MARSEILLE PROVENCE AP, France

WMO: 076500

Lat: 43.438N

Lon: 5.216E

Elev: 74

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.0	(c) 30.6	(d) 17.3	(e) 13.2	(f) 33.9	(g) 21.0	(h) 15.8	(i) 35.8	(j) 40.1	(k) 46.3	(l) 35.3	(m) 46.4	(n) 6.3	(o) 140	(p) 0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.9	(c) 91.6	(d) 69.7	(e) 89.3	(f) 69.3	(g) 87.2	(h) 68.8	(i) 73.8	(j) 84.7	(k) 72.5	(l) 83.6	(m) 71.2	(n) 82.4	(o) 13.2	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	112.1	79.5	68.8	106.0	78.4	67.3	100.6	77.7	37.3	85.0	36.1	83.7	35.0	82.2	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 35.4	(b) 30.8	(c) 26.9	(e) 24.3	(f) 96.0	(g) 3.2	(h) 3.0	(i) 21.9	(j) 98.2	(k) 20.1	(l) 99.9	(m) 18.2	(n) 101.6	(o) 15.9	(p) 103.8		
			DB													
			WB													
				22.6	76.0	3.3	1.3	20.2	77.0	18.3	77.7	16.5	78.5	14.1	79.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	45.6	46.6	52.4	57.7	64.9	72.6	77.3	76.5	69.3	62.6	52.7	46.9
	DBStd	12.46	5.94	5.90	4.99	4.65	4.80	5.24	4.11	4.02	4.74	5.65	6.64	6.35
	HDD50	504	161	121	33	3	0	0	0	0	0	2	47	137
	HDD65	2858	602	516	392	222	60	5	1	0	16	111	370	563
	CDD50	4338	24	24	107	235	461	679	846	823	578	394	128	39
	CDD65	1215	0	0	0	3	56	234	382	358	143	38	1	0
	CDH74	10138	0	0	1	27	360	1954	3651	3192	844	109	0	0
	CDH80	3488	0	0	0	1	58	635	1437	1186	166	5	0	0
(14) Wind	WSAvg	10.8	10.2	10.9	11.3	11.6	11.2	11.0	11.6	10.6	10.4	9.9	10.7	9.8
(15) Precipitation	PrecAvg	19.7	1.8	1.3	1.0	1.9	1.5	1.0	0.4	0.9	3.0	2.8	2.6	1.6
	PrecMax	28.9	5.4	4.6	2.8	5.4	3.5	4.3	2.6	2.9	8.3	5.6	6.4	5.9
	PrecMin	9.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.9	0.2	0.0
	PrecStd	5.4	1.5	1.2	0.8	1.3	1.1	1.1	0.6	0.8	2.2	1.3	1.7	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	64.2	71.2	77.3	85.8	93.4	94.4	95.3	87.5	80.2	70.1	63.6
		MCWB	53.5	53.0	55.8	60.6	65.7	70.1	69.9	70.7	67.8	66.1	61.4	56.6
	2%	DB	59.2	60.9	67.2	73.3	80.9	89.7	91.5	91.3	84.4	77.0	67.1	61.0
		MCWB	52.5	51.5	54.4	59.1	63.0	69.0	69.7	70.0	67.5	65.3	60.0	55.2
	5%	DB	57.1	58.4	64.3	70.2	77.9	86.4	89.5	88.7	81.6	74.4	65.1	59.0
		MCWB	51.6	50.5	53.5	56.8	61.9	68.0	69.1	69.7	66.4	64.2	58.7	53.7
	10%	DB	55.2	56.2	61.9	67.5	75.1	83.7	87.4	86.3	78.9	72.0	63.0	56.9
		MCWB	49.9	49.5	52.3	55.5	60.9	66.7	68.8	69.0	65.3	63.0	57.3	52.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	54.8	57.9	62.6	68.2	73.9	75.5	75.5	72.3	68.9	63.4	58.2
		MCDB	59.7	60.7	67.0	74.0	80.4	87.9	85.2	85.7	80.7	76.0	67.8	62.0
	2%	WB	53.8	53.2	56.1	60.2	65.5	71.7	73.7	74.0	70.5	67.2	61.2	56.1
		MCDB	58.1	58.8	64.0	70.6	76.6	84.4	84.9	84.4	79.1	74.3	65.8	59.9
	5%	WB	52.1	51.8	54.8	58.5	63.8	70.0	72.3	72.7	69.0	65.7	59.5	54.4
		MCDB	56.5	57.2	62.1	67.8	74.0	82.4	84.0	83.4	77.5	72.6	64.3	58.2
	10%	WB	50.4	50.1	53.4	56.9	62.3	68.3	70.9	71.4	67.5	64.0	57.8	52.6
		MCDB	54.7	55.3	60.4	65.1	72.1	80.6	82.9	82.1	76.1	70.4	62.6	56.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.8	17.2	17.0	17.6	18.5	18.9	18.8	17.4	15.6	14.4	13.9
		MCDBR	14.7	18.5	20.7	20.8	21.4	21.5	21.2	21.7	20.4	17.3	14.8	14.1
		MCWBR	10.0	11.7	11.5	10.6	9.7	8.7	8.5	9.1	9.4	9.1	9.4	9.6
	5% WB	MCDBR	13.2	16.3	18.0	18.9	19.0	19.6	18.9	18.8	17.1	14.9	13.2	12.3
		MCWBR	9.8	11.1	10.8	10.6	9.6	8.9	8.9	9.5	9.2	8.7	9.2	9.2
(40) Clear-Sky Solar Irradiance	taub		0.321	0.338	0.372	0.396	0.406	0.405	0.399	0.397	0.386	0.380	0.345	0.317
	taud		2.493	2.444	2.367	2.317	2.335	2.363	2.382	2.392	2.410	2.433	2.476	2.507
	Ebn at Noon		258	272	274	276	275	275	272	272	266	252	245	248
	Edh at Noon		23	28	34	39	39	38	37	36	32	28	23	21
(44) All-Sky Solar Radiation	RadAvg		574	882	1308	1708	2061	2373	2369	2036	1547	982	631	498
	RadStd		58	84	97	112	142	92	85	62	84	82	69	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	+0.54	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	N/A	+90

Nomenclature: See separate page